UTSAV CHAUDHARY

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Senior **Full Stack Engineer** with **7+ years** of hands-on **backend** and **frontend** development, plus **AI agents**, **bots**, and **LLM** systems, building production applications in **regulated utility**, **energy**, and **enterprise consulting** environments. Expert in **Python**, **TypeScript**, and **JavaScript** with **Angular**, **React.js**, and **Next.js**, and in **REST APIs** using **FastAPI**, **Flask**, and **Node.js**. Delivers **solution design**, development quality, and delivery execution on **AWS** and **Azure** (compute, storage, networking, platform services), and guides teams on **CI/CD**, **Git workflows**, **Terraform**, and **GitHub Actions** with consistent **SOLID**, **automated testing**, **coding standards**, and **secure coding**.

Technical Skills

Languages & Frontend: Python, TypeScript, JavaScript, C#, Angular, React.js, Next.js, Redux, TailwindCSS, D3.js, Chart.js

Backend & APIs: FastAPI, Flask, Django, ASP.NET, ASP.NET Web API, Node.js (Express), Pydantic, REST APIs, OpenAPI, Swagger, gRPC, WebSockets, GraphQL

AI, LLMs, Agents, & Bots: Azure OpenAI, GPT-4o, GPT-4o-mini, Llama, Hugging Face, RAG, embeddings, vector search, MCP servers, AI agents, Ollama, Whisper, TensorFlow Lite, prompt engineering

Cloud, DevOps, & CI/CD: AWS (EC2, EKS, Lambda, RDS, S3, VPC, IAM, API Gateway, CloudWatch), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Key Vault), Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, Bitbucket Pipelines, Git, GitFlow, environment promotion

Networking & Security: VNets, Security Groups, NSGs, DNS, VPN, LDAP, RBAC, SSL, TLS, OAuth 2.0, NERC CIP, ISO 27001, PCI, OWASP, secure coding, vulnerability assessments, enterprise risk controls

Engineering Practices & Tools: OOAD, SOLID, automated testing, code reviews, design reviews, branching strategies, PostgreSQL, MongoDB, Redis, Kafka, Jira

Experience

Sr. Full Stack Engineer / Technical Lead, Hybrid, **MidAmerican Energy**, TX — Feb 2025 to Present

- Led parallel **AI** and product workstreams for a regulated electric and natural gas utility, turning requirements into technical specifications for **APIs** and UI across grid telemetry and smart meter (AMI) services.
- Led code and design reviews keeping workstreams consistent on **OOAD**, **SOLID**, automated testing, coding standards, and documentation.
- Designed **AI agents, bots, and MCP servers** with custom tools and APIs; owned the manager agent routing work to specialist agents with human-in-the-loop gates for outage management and metering.
- Built the shared **Python/TypeScript** tool layer (search, scrape, DB, Gmail) with schemas and retries, wiring **Azure OpenAI GPT-4o** and a Hugging Face model as callable tools with NERC CIP document retrieval into MongoDB.
- Built **React.js/Next.js TypeScript** apps with TailwindCSS, including an Admin Portal for KPI, agent-run, outage, and AMI metering metrics; Figma to production UI with RBAC.
- Developed **Python** microservices and **Node.js REST APIs** over MongoDB and Firebase, integrating ERP, customer information system (CIS), and market data feeds with PCI-aligned secure coding for customer payments.
- Partnered with DevOps on **CI/CD** (GitHub Actions, Bitbucket), Terraform, Docker on EKS/AKS, and environment promotion; led root cause analysis with Prometheus and CloudWatch.

Full Stack Engineer, On-Site, **Chevron**, GA — Jan 2024 to Jan 2025

- Implemented and reviewed **Python FastAPI/Flask** and **Node.js REST APIs** for drilling reporting and refinery maintenance workflows, with automated testing before **AKS** promotion, helping keep 99.9% availability behind Application Gateway and Front Door.
- Delivered **Angular** and **React.js** TypeScript features for field operations and maintenance workflow portals with real-time telemetry views, keeping UI routing and API contracts intact through environment promotion on Azure.
- Performed code and design reviews on portal and API changes; applied OOAD, SOLID, and OWASP with coding standards and secure coding, using **Azure Key Vault** to protect proprietary seismic and operational data in Azure SQL.
- Helped migrate the portal and telemetry stack (Angular, React, Python, Node.js) from legacy hosting to cloud-native **AKS** on Azure, improving resource utilization by 40% while preserving service boundaries and API contracts.
- Supported canary releases and environment promotion through **Azure DevOps** pipelines, Terraform, Git workflows, and branching strategies, contributing to a 75% reduction in deployment related incidents.
- Contributed to monitoring with Application Insights, Azure Monitor, and Grafana, supporting MTTD from 30 minutes to 5 minutes; led root cause analysis and vulnerability assessments with Risk and Compliance.

Full Stack Engineer CO-OP, On-Site, **Midh Technologies**, TX — June 2023 to Dec 2023

- Built **Python FastAPI** and **Flask REST APIs** for **Llama** and Hugging Face inference behind AWS API Gateway / Azure API Management, with IAM authentication, request throttling, and Pydantic validation.

- Developed **React.js** analytics dashboards with TypeScript, D3.js, and granular RBAC for internal teams and client stakeholders, translating analyst requirements into production UI workflows.
- Fine-tuned **Llama** models on GPU-backed **Kubernetes** (EKS/AKS) clusters tuned with node pools, resource quotas, and auto scaling; contributed Python ETL and Pandas jobs that loaded curated tables into PostgreSQL and MongoDB for dashboard and API consumers.
- Built and deployed the company web platform in **Next.js** and React with secure RBAC and scalable API integrations on Azure App Service and AWS Amplify.
- Built production **CI/CD** with GitHub Actions and Azure DevOps: pytest unit tests, Docker image builds, security scanning, Terraform, and environment promotion.

Software Engineer, Python, On-Site, **Accenture**, India June 2021 to July 2022
- Built a responsive **React** frontend on a **FastAPI** backend for enterprise client accounts in a global consulting engagement, delivering interactive D3.js and Chart.js metrics dashboards that improved data visualization efficiency by 45%.
- Optimized **Python FastAPI REST APIs** with Pydantic models, sustaining 250ms average response time behind Nginx with SSL/TLS termination on hybrid on premise deployments.
- Built **Python** scraping/ETL with Selenium, BeautifulSoup, and Pandas, cutting manual collection time by 60%; added WebSocket streaming so operational dashboards stayed current.
- Implemented self hosted **CI/CD** with GitHub Actions and Jenkins, Git branching strategies, and Prometheus/Grafana observability; applied code reviews and ISO 27001-aligned secure coding, reducing operational overhead by 60%.

Software Engineer, Remote, **VRoKCs**, MO March 2020 to May 2021
- Deployed **Python Flask** web applications and REST APIs on **Azure App Services** with high availability, auto scaling, and integrated monitoring across compute, storage, and networking.
- Built a **Flask** app (Python, JavaScript) with TensorFlow Lite for real time, in browser sign language model evaluation via image upload, webcam streaming, and live prediction display.
- Built **CI/CD** through Azure DevOps and Git versioning for model and application deployment, using Azure Blob Storage for artifacts and Docker for inference services, reducing deployment cycles by 40%.
- Applied automated testing and secure coding on critical path Flask APIs and JS client work; partnered with ML engineers on Python preprocessing for data collection, annotation, and dataset preparation.

Software Developer, On-Site, **Wonderbiz.in**, India March 2019 to Feb 2020
- Implemented **REST APIs** in **Python Flask** and **Node.js** for Industrial IoT sensor ingestion, asset tracking, and dashboard services, with the customer facing **Angular** analytics dashboard and JavaScript UI built against them.
- Used Git versioning, Jenkins builds, and release checklists for environment promotion across Dev, Test, and Production before deploying Industry 4.0 products to customer environments.
- Performed code reviews, unit and integration automated testing, and secure coding checks (input validation, auth, logging) before production releases.

Web/Game Developer, On-Site, **underDogs Gaming Studios**, India Jan 2018 to Feb 2019
- Built game service **REST APIs** in **Python** and **Node.js**, integrating the Quarters API for wallets, currency conversion, and secure transactions across multiple titles.
- Wrote **Python** tooling and JavaScript admin screens for payments, inventory, and shipping, keeping client/server flows reliable when requests failed or the network dropped.
- Applied API security, logging, and integration automated testing, and stood up release and hosting for game backends with Git source control and health checks.

Education

State University of New York at Binghamton Aug 2022 to Dec 2023
Master of Science in Computer Science 3.4/4.00 GPA

Certifications

Microsoft Certified: DevOps Engineer Expert View Credential
AZ-400: Designing and Implementing Microsoft DevOps Solutions
AZ-900: Microsoft Certified: Azure Fundamentals
Google IT Automation with Python (Specialization) View Credential